Product Requirement Document (PRD)

Personal Health Record Management and Sharing System (PHRMS)

Course: MIS 4500 – Technology Capstone Project (Fall 2026)

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1. Executive Summary & Product Vision

Patients frequently receive medical reports, prescriptions, lab results, and consultation notes from multiple healthcare providers. Keeping these fragmented records organized and retrieving relevant history during a short medical consultation is challenging.

The **Personal Health Record Management and Sharing System (PHRMS)** is a web-based academic prototype designed to centralize personal health records through a structured, three-role workflow:

1. **Patients** upload medical documents, enter health data, organize approved records chronologically, generate consultation-ready summaries, and control which doctors can view their profile.
2. **Document Review Team Members** administratively inspect submitted documents for completeness and legibility—approving, rejecting, or requesting corrections before records enter the official patient timeline.
3. **Doctors** view only the chronological records and consultation summaries explicitly authorized by the patient.

2. Target User Personas

Persona	Role Code	Primary Goals	Key Pain Points Addressed
Proactive Patient	`PATIENT`	Keep all medical records in one place, track administrative approval, prepare concise summaries before appointments, and control privacy.	Lost paper reports, disorganized PDFs, repeating medical history to new doctors, and lack of visibility into who can see their records.
Administrative Reviewer	`REVIEWER`	Verify that uploaded documents match entered metadata (date, category, provider, legibility) and provide clear feedback when corrections are needed.	Unstructured, mislabeled, or incomplete patient uploads cluttering clinical views.
Consulting Physician	`DOCTOR`	Quickly review a patient's verified chronological history and pre-selected consultation summary during an appointment.	Digging through unverified, unsorted files or lacking context during a consultation.

3. Functional Requirements & User Stories

3.1 Role-Based Authentication (FR01)

- **Requirement:** The system shall allow users to log in according to their assigned roles (`PATIENT`, `REVIEWER`, `DOCTOR`) and redirect them to their respective interfaces.
- **Acceptance Criteria:**
  - Unauthenticated users are redirected to the login screen.
  - Attempting to access a route belonging to another role returns an access-denied message or redirects to the user's home dashboard.
  - Demo credentials for fictitious patients, reviewers, and doctors are pre-seeded for local testing.

3.2 Medical Document Submission & Health Data Entry (FR02)

- **Requirement:** Patients shall be able to upload medical documents (PDF, PNG, JPG) and enter structured health information (Title, Document Type, Healthcare Provider, Document Date, Clinical Notes/Vitals).
- **Acceptance Criteria:**

- New submissions automatically receive the status `Pending Review`.
- Patients can also re-submit/edit records that have been marked `Correction Requested` by the review team.

3.3 Administrative Document Review Workflow (FR03)

- **Requirement:** The Document Review Team shall be able to inspect submitted documents and **Approve**, **Reject**, or **Request Corrections** with mandatory reviewer notes for non-approved items.
- **Acceptance Criteria:**
  - Reviewer dashboard lists all `Pending Review` submissions in a queue.
  - Reviewer can preview the uploaded file alongside patient-entered metadata.
  - Selecting `Approved` moves the record into the patient's chronological timeline.
  - Selecting `Correction Requested` or `Rejected` records the reviewer's comment and updates the patient's submission tracker.

3.4 Submission Status Tracking (FR04)

- **Requirement:** The system shall display the current review status of all submitted documents to the patient.
- **Acceptance Criteria:**
  - Statuses are visually distinguished using the medical color palette (`Approved`, `Pending Review`, `Correction Requested`, `Rejected`).
  - Reviewer feedback comments are clearly visible to the patient on any `Correction Requested` or `Rejected` item.

3.5 Chronological Health Record Profile (FR05)

- **Requirement:** Approved records shall be automatically organized chronologically (by the medical document's date) in the patient's health profile.
- **Acceptance Criteria:**
  - Only `Approved` records appear on the official chronological timeline.
  - Timeline supports filtering by Document Category (e.g., Lab Report, Prescription, Imaging, Consultation Note, Vaccination/Vitals).

3.6 Patient-Controlled Doctor Access (FR06)

- **Requirement:** Patients shall be able to authorize or revoke specific doctors' access to their health records at any time.
- **Acceptance Criteria:**
  - Patient can view a list of registered doctors and toggle access (`Grant Access` / `Revoke Access`).

  - Optional access scope notes can be viewed.
  - Revoking access immediately blocks the doctor from viewing the patient's records or summaries.

3.7 Authorized Doctor View (FR07)

- **Requirement:** Doctors shall be able to view only the records and consultation summaries of patients who have actively authorized them.
- **Acceptance Criteria:**
  - Doctor dashboard lists only currently authorized patients.
  - Attempting to load a non-authorized patient's profile directly is blocked and logged.

3.8 Consultation-Ready Summary Generator (FR08)

- **Requirement:** The system shall generate a structured, consultation-ready summary from a patient-selected subset of approved records.
- **Acceptance Criteria:**
  - Patient selects specific approved records via checkboxes, adds a "Reason for Visit / Chief Complaint" note, and generates a Consultation Summary.
  - The summary compiles patient demographics, key vitals/allergies, reason for visit, and the selected chronological records into a clean, printable/viewable consultation sheet visible to authorized doctors.

3.9 Activity & Audit Logging (FR09)

- **Requirement:** The system shall record relevant access, submission, and review actions in an audit log.
- **Acceptance Criteria:**
  - Logs capture timestamp, actor (user & role), action type, and target entity.
  - Patients can view an "Access & Activity History" tab showing when reviewers or doctors accessed their records.

4. Non-Functional Requirements (NFR01–NFR07)

ID	Category	Specification
NFR01	**Usability**	Interfaces must be clean, consistent, and intuitive across all three roles, utilizing clear status badges and accessible typography.
NFR02	**Security**	Enforce session authentication and strict server-side role-based access control (RBAC) checks on all routes.
NFR03	**Privacy**	Doctors can only query records where an active authorization link exists between the doctor and patient.
NFR04	**Reliability**	State transitions (`Pending` $\rightarrow$ `Approved` / `Correction Requested` / `Rejected`) must be atomic and consistent.
NFR05	**Performance**	Local page loads, timeline queries, and summary generation must execute in < 500 ms under normal prototype usage.
NFR06	**Maintainability**	Modular codebase separating data models, route controllers, and UI templates.
NFR07	**Data Integrity**	Approved records cannot be silently altered without re-triggering administrative review; unauthorized modifications are prevented.

5. Scope Boundaries & Assumptions

- **In Scope:** Local web execution, three distinct role interfaces, document upload & metadata entry, administrative review workflow, chronological health timeline, granular doctor access toggle, consultation summary builder, audit logging, and pre-populated fictitious test data.
- **Out of Scope:** Clinical diagnosis or automated medical decision support, cryptographic or third-party verification of hospital document authenticity, real patient data (PII/PHI), cloud deployment, or live hospital EHR integration.